Abdul Monaf Chowdhury

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EDUCATION

UNIVERSITY OF DHAKA, BANGLADESH ***Jan 2019 – Dec 2023***
Bachelor of Science in Robotics & Mechatronics Engineering
CGPA: 3.87/4.00

RELEVANT COURSEWORK: Introduction to Machine Learning, Artificial Intelligence, Digital Image Processing, Digital Signal Processing, Object Oriented Programming, Human Robot Interaction, Advanced Robotics, Electrical Machines, Calculus, Linear Algebra, Statistics

SKILLS: Software: C, C++, Python, MATLAB, PyTorch, TensorFlow
Language: Fluent in both English and Bangla

RESEARCH EXPERIENCE

MAIM LAB, UNIVERSITY OF DHAKA, BANGLADESH ***Feb 2024 – Present***
Research Assistant

- Collaborated with University College Dublin researchers on 'Wellcome Leap' funded project titled 'Translation of a Wearable Fetal Movement Monitor towards Stillbirth Prevention' to develop a wearable device to monitor fetal movements to prevent stillbirth.
- Formed different thresholding pipelines to identify optimal thresholding mechanism for ideal feature extraction from sensor data
- Devised and optimized both machine learning and deep learning systems to streamline the predictions based on extracted features of sensor data and improved maternal sensation detection.

UNIVERSITY OF DHAKA, BANGLADESH [Link] ***Jan 2023 – Dec 2023***
Undergraduate Researcher

- Studied, analyzed, and catalogued ideas from prior state of the art research papers on top level Computer Vision conferences and journals
- Collaborated with graduate students and generated different notions and tested them on several UAV action recognition datasets
- Initiated, designed, and built a deep learning system incorporating Fast Fourier Transforms to classify and improve drone based human action recognition for surveillance for undergraduate thesis project

TEACHING EXPERIENCE

TUTOR ***Mar 2022 – Nov 2022***

- Assisted and guided 12th grade students one on one in Calculus, Fundamental Physics, and Introductory C programming

PUBLICATIONS

- "FFT-UAVNet: FFT Based Human Action Recognition for Drone Surveillance System" [Link]
  Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan
  5th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI), 2023, In Press

PROJECTS

PROXEMICS AND SOCIAL INTERACTION PATTERNS IN ASD CHILDREN ***Sep 2023 – Present***

Human Robot Interaction

- Formulated a YOLOv8 based system to determine the ideal proxemics of autism spectrum disorder (ASD) children
- Examined and analyzed the behavioural responses of children diagnosed with ASD in the presence of specific actions performed by the NAO robot.

FINANCIAL DATA ANALYSIS [Link] ***Aug 2022 – Nov 2022***

Reinforcement Learning

- Implemented Approximate Q Learning for three Bangladeshi stocks to generate Buy, Sell, and Hold orders
- Achieved 11% return of investment for the 3 stocks beating the DSE 30 index

AWARDS & HONORS

- Secured 5th place in Dataverse Challenge - ITVerse, Bangladesh, *2023* [Link]
- Became 1st Runner-up in Intra-Department Soccer Bot Championship, University of Dhaka, *2019*

WORK EXPERIENCE & LEADERSHIP

RMEDU STUDENT CLUB, UNIVERSITY OF DHAKA ***Mar 2022 – Feb 2024***

General Secretary

- Successfully organized and supervised frequent cultural events, sports events, and competitions
- Arranged and delegated paper reading sessions, workshops, training sessions
- Addressed numerous concerns and issues of the student body and issued relevant responses

BANGLADESH ROBOTICS OLYMPIAD ***Sep 2019 – Aug 2022***

Academic Team Mentor

- Developed questions for National Robotics Olympiad and organized workshops
- Helped materialize the National Robotics Olympiad for 4 years

IEEE ROBOTICS & AUTOMATION SOCIETY, UNIVERSITY OF DHAKA ***Jul 2021 – Jun 2022***

Program Co-Ordinator

- Directed and facilitated several webinars, interactive sessions, and expert talks.
- Collaborated with other IEEE societies across the country and accelerated IEEE RASDU membership by 15%